Neural Feedback Optimization Theory (NFOT):

The Write-Back Gap as a Trial-Level Temporal Variable in Closed-Loop Learning Systems

Folarera Ayobami Kassim
Independent Researcher
folarera.kassim@gmail.com
ORCID: 0009-0007-2148-5675

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Note on versions. This is v5 of this preprint. Previous versions (v1, v2) contained a specific numerical claim—a characteristic timescale of approximately 28.42 ms—derived from a simulation whose assumptions could not be independently verified, together with a derived functional form (the Lorentzian efficiency law). Both have been removed. v3 stated the hypothesis and a protocol for testing it, without data. v4 added a preliminary observational analysis of 330 trials from bigP3BCI Study M. The present version extends that analysis to 444 trials from 15 subjects and reports support for the distributional prediction.

Abstract

Closed-loop neuroadaptive systems—including brain-computer interfaces (BCIs), neurofeedback systems, and adaptive learning technologies—detect a biological or behavioral event, compute an intervention, and deliver feedback. The elapsed time between the event and the delivered feedback is usually treated as an engineering property to be minimized. This paper argues that it should be treated as a scientific variable. I define the *Write-Back Gap* L as the achieved, trial-level interval between a prespecified biological event t_0 and the completion of the corresponding feedback t_{fb}, measured on matched pairs. I argue this differs from ordinary latency in four specific ways, and that the *distribution* of L—not only its mean—may carry information about learning outcomes.

The paper states a falsifiable predictive hypothesis and a stronger causal hypothesis, presents a preregistered experimental protocol (PTSP) for testing both, and specifies in advance what results would disconfirm them. It also surveys publicly available closed-loop datasets to determine whether the hypothesis can currently be tested. No functional form is assumed, no numerical time constant is claimed, and no simulation output is presented as a biological finding. Preliminary observational analysis of 444 trials from bigP3BCI Study M supports the predictive hypothesis at the trial level ($\beta = -0.111$, $p = 0.030$; leave-one-subject-out 15/15 negative) and supports the distributional prediction: latency variance carries information about accuracy beyond the mean (partial correlation $p = 0.027$; out-of-sample AUC improvement $+0.076$, 95% CI $[+0.021, +0.131]$). Whether the causal prediction survives is an empirical question.

1 Introduction

Closed-loop systems that adapt to a user's biological or behavioral state share a common structure: a target event occurs, the system detects it, computes a response, and delivers feedback. The time this takes is normally treated as an engineering cost. Decoders are benchmarked on latency; architectures are optimized to reduce it; a faster pipeline is assumed to be a better one.

This paper argues that this framing misses a scientific question. Between the target event and the delivered feedback, the biological state that determines whether learning occurs may change. Synaptic eligibility traces decay. Neuromodulatory signals rise and fall. The brain does not wait for the pipeline to finish. If the temporal relationship between an event and its feedback affects learning, then that relationship is a variable worth measuring—not merely an engineering delay to be minimized.

The paper does five things. Section 2 defines the variable, called the Write-Back Gap, and distinguishes it from ordinary latency. Section 3 states the hypothesis under which it matters and specifies what would falsify it. Section 4 presents the strongest testable prediction. Section 5 surveys public datasets, and Section 6 reports a preliminary observational analysis from one of them. Section 7 presents a protocol for testing the causal claim.

2 The Write-Back Gap

2.1 Definition

Let t_0 denote the instant at which a prespecified target biological state transition occurs. Depending on the system, t_0 may correspond to a pre-/postsynaptic spike relationship, a decodable transition in a neural population trajectory, a behavioral or physiological event, or a crossing of a prespecified phase ϕ^* of an ongoing electrophysiological oscillation.

Let t_{fb} denote the instant at which the corresponding adaptive feedback is fully delivered to the biological substrate. The Write-Back Gap is

$$L \equiv t_{\mathrm{fb}} - t_0, \qquad L \geq 0. \tag{1}$$

The total gap decomposes as

$$\begin{aligned} L = {} & L_{\mathrm{acq}} + L_{\mathrm{filter}} + L_{\mathrm{decode}} + L_{\mathrm{compute}} \\ & + L_{\mathrm{network}} + L_{\mathrm{act}} + L_{\mathrm{other}}. \end{aligned} \tag{2}$$

Each component can be instrumented separately, while the biological experiment operates on the achieved end-to-end value.

2.2 Why This Is Not Simply Latency

The distinction is not terminological. It consists of four commitments.

1. **Relative to what.** L is measured relative to a *biologically defined* event t_0, not an arbitrary system timestamp. The choice of t_0 is a scientific commitment and must be made independently of the observed learning outcome.

2. **Measured at which endpoint.** L terminates at the completion of feedback delivery to the substrate, not at the moment feedback is enqueued or dispatched. Software-side timestamps and delivered-feedback timestamps can differ substantially.

3. **Correspondence.** L is defined on matched (t_0, t_{fb}) pairs. A mean latency over a session is not the same object as a per-trial gap between a specific event and the feedback generated in response to it.

4. **Treatment.** Conventional latency analysis treats latency as an engineering property to be minimized. Here L is a variable whose relationship to learning is an empirical question, whose distribution may itself be predictive, and whose connection to biological eligibility is a hypothesis rather than an assumption.

If these four commitments are not upheld, L collapses into ordinary system latency and the contribution of this paper disappears.

3 The Core Hypothesis and Its Falsification

3.1 The Hypothesis

Predictive hypothesis. After the target event and feedback modality are prespecified, the achieved trial-level Write-Back Gap L contains reproducible out-of-sample information about learning-related outcomes under at least some closed-loop conditions.

Causal hypothesis (stronger, tested separately). Manipulating L while holding relevant feedback properties constant changes the learning-related outcome.

The predictive and causal claims are not interchangeable. The predictive claim can be supported by observational data; the causal claim requires controlled manipulation.

3.2 What Would Falsify It

A framework that cannot be disconfirmed by any conceivable result is not scientific. Table 1 states, in advance, how a range of possible outcomes bears on the hypothesis.

Table 1: Null-result interpretations, stated in advance.

Empirical result	**Interpretation**
L improves preregistered out-of-sample prediction	Supports the predictive hypothesis
L adds no reproducible predictive information after controls	Challenges the predictive hypothesis; the framework should be abandoned as a distinct contribution
Mean L does not matter, but variance or tails do	Supports the distributional prediction while weakening a mean-only reading
Manipulating L does not change the outcome under matched feedback content	Challenges the causal hypothesis

4 The Distributional Prediction

This is the strongest testable claim in the paper.

4.1 Statement

Consider two closed-loop systems A and B with

$$\mathbb{E}[L_A] = \mathbb{E}[L_B] \qquad \text{but} \qquad \mathrm{Var}(L_A) \neq \mathrm{Var}(L_B), \tag{3}$$

and with matched feedback content, modality, and task. The prediction is that learning outcomes may differ between A and B.

4.2 Why Mean Latency Alone May Be Insufficient

If learning depends on the residual eligibility available at the moment of delivery, and if eligibility decays nonlinearly, then a system with matched mean but higher variance delivers some feedback very early (while eligibility is high) and some very late (while eligibility has largely decayed). The late trials contribute disproportionately little to learning. A mean-only summary cannot capture this asymmetry. The prediction is therefore not a background observation that latency varies; it is a specific claim that the *shape* of the latency distribution carries information that the mean does not.

4.3 How to Test It

To render the prediction falsifiable, a study must:

1. prespecify the distributional summary whose effect is tested (variance, a tail quantile, or a tail-risk functional);
2. construct conditions with matched means but different distributional summaries;
3. test whether the distributional summary predicts outcome after controlling for the mean;
4. preregister the predicted direction.

Without these commitments, "distribution matters" risks becoming unfalsifiable, because almost any latency statistic can be examined after the fact.

5 Dataset Survey

To determine whether the hypothesis can currently be tested, I surveyed publicly available closed-loop datasets. I searched OpenNeuro for keywords including "neurofeedback," "P300 speller," and "BCI," and examined the most relevant candidates for whether they preserved trial-level selection or feedback timing.

5.1 BCI-FIT (OpenNeuro ds007720)

BCI-FIT is a communication BCI dataset from five participants with ALS, collected with BciPy [7]. It includes calibration and copy-spelling tasks in BIDS format.

I examined the CopyPhrase task files for participant `sub-P1`, session `ses-2`, run 7. The event files are:

- `sub-P1_ses-2_task-MatrixCopyPhrase_run-7_events.tsv` — contains `onset`, `duration`, `trial_type`, `value`, `sample`. The `trial_type` column contains only `target` and `nontarget` stimulus labels. No selection or feedback events are recorded.
- `sub-P1_ses-2_task-MatrixCopyPhrase_run-7_eeg.vmrk` — contains only `Stimulus` markers. No `Response` or selection markers.

The selection events—the moments at which the system made a character decision and delivered feedback—are not present in the published BIDS files.

5.2 Chandravadia et al. (OpenNeuro ds005028)

This dataset compares P300 flashing paradigms in online typing with language models [1]. It contains raw EEG in `.dat` and `.edf` formats. No `events.tsv`, `.vmrk`, or behavioral event files are included. Event information, if preserved, is stored inside the raw BCI2000 binary files and is not directly accessible.

5.3 bigP3BCI Study M

In contrast to the two datasets above, the adaptive diffuse condition of bigP3BCI Study M (PhysioNet) preserves trial-level selection and feedback events in its raw EDF files. Event channels including `StimulusBegin`, `DisplayResults`, `CurrentTarget`, and `SelectedTarget` are recorded at full sampling rate.

I wrote an extraction pipeline (VampireBCI) that reads these channels, decodes the selected cell using the grid encoding cell $= (\text{row} - 1) \times 8 + \text{col}$ (verified on 4192/4192 samples with zero exceptions), identifies the feedback onset from `DisplayResults` rising edges, and computes L and Y on matched (t_0, t_{fb}) pairs. Extraction from 74 EDF files yielded 444 trials from 15 subjects. The event channels therefore contain the information required to test the predictive hypothesis, and the extracted dataset is used for the preliminary analysis in Section 6.

5.4 Finding

BCI-FIT and Chandravadia et al. do not preserve trial-level selection or feedback timing in an accessible format. bigP3BCI Study M does. The data required to test the Write-Back Gap hypothesis is therefore not routinely recorded across the field, but it exists in at least one public dataset, and the analysis below uses it.

6 Preliminary Observational Evidence

The predictive hypothesis (Section 3) does not require a controlled manipulation; it can be tested against observational data in which L varies naturally. The adaptive diffuse condition of bigP3BCI Study M provides such data.

6.1 Dataset

Extraction from 74 EDF files yielded 444 trials from 15 subjects (M_04–M_20, excluding M_08 which does not appear in the dataset). Overall accuracy was 82.9% (368/444). The Write-Back Gap had mean $L = 7.06$ s and standard deviation 3.32 s, with range 1.88–15.0 s.

6.2 Trial-Level Regressions

Four logistic models were fit to the trial-level outcome Y:

Table 2: Trial-level logistic regressions of Y on L.

Model	$\boldsymbol{\beta_L}$	$\boldsymbol{p}$
Pooled	-0.199	< 0.0001
Subject fixed effects	-0.094	0.049
Mixed model (random intercept)	-0.024	0.0003
Within-subject (centered L)	-0.111	0.030

The effect is negative in every specification. The within-subject coefficient ($\beta = -0.111$, $p = 0.030$) is smaller than the estimate from an earlier 11-subject subset ($\beta = -0.180$, $p = 0.0023$); the effect is real but modest.

6.3 Out-of-Sample Validation

Leave-one-subject-out cross-validation was performed on the within-subject model. All 15 held-out subjects yielded a negative L coefficient (mean $\beta = -0.1109$, standard deviation 0.0195). The out-of-sample coefficient was indistinguishable from the in-sample estimate ($\beta = -0.1111$).

6.4 Tail Analysis

The effect is concentrated in the tail of the L distribution. Splitting at the 75th percentile ($L = 8.88$ s), trials in the top quartile were 67.9% accurate versus 88.0% for the rest (Fisher exact $p < 0.0001$). At the 90th percentile ($L = 12.00$ s), 63.0% versus 85.2% ($p = 0.0006$). A threshold model at 8.88 s yields $\beta = -1.24$ ($p < 0.0001$), odds ratio 0.289.

6.5 Distributional Prediction

Section 4 predicts that the distribution of L, not only its mean, carries information about learning outcomes. Three tests bear on this claim.

Between-subject correlations. Across the 15 subjects, $r(L_{\text{mean}}, \text{acc}) = -0.759$ ($p = 0.0010$) and $r(L_{\text{std}}, \text{acc}) = -0.850$ ($p = 0.0001$). Both survive Bonferroni correction for two tests (threshold $p < 0.025$).

Partial correlation. In a multiple regression with both predictors, L_{std} remains significant ($\beta = -0.135$, $p = 0.027$) while L_{mean} does not ($p = 0.951$). The variance of L carries information beyond the mean.

Out-of-sample model comparison. Leave-one-subject-out prediction was compared across three models: mean-only (AUC = 0.656), mean plus L^2 (AUC = 0.655), and mean plus subject-level L_{std} (AUC = 0.731, log-loss 0.408 vs. 0.435 for mean-only). A bootstrap of the AUC difference (A vs. C) gave a mean of +0.076 with 95% CI $[+0.021, +0.131]$ and $P(\text{diff} > 0) = 0.996$.

6.6 Interpretation

The predictive hypothesis is supported. Trial-level L contains reproducible out-of-sample information about accuracy; the effect is stable across all 15 held-out subjects and is concentrated in the tail of the L distribution. The effect is modest when summarized by a single linear coefficient, but the tail analysis and the threshold model show that late feedback is disproportionately associated with incorrect selections.

The distributional prediction is supported. All three tests agree: L_{std} correlates more strongly with accuracy than L_{mean}; it remains significant after controlling for L_{mean}; and adding it improves out-of-sample prediction with a confidence interval that excludes zero. The shape of the L distribution carries information that the mean does not.

The causal hypothesis is not tested here. It requires the controlled manipulation specified in Section 7.

6.7 Scope Conditions

Five of fifteen subjects (M_10, M_12, M_14, M_17, and M_18) performed at or near ceiling ($\geq 96.7\%$ accuracy), contributing little outcome variance. The trial-level effect is estimated primarily from the subjects with sufficient error rates. The hypothesis as stated applies "under at least some closed-loop conditions"; the condition observed here is one in which participants are still making errors. Subjects at ceiling provide no errors to predict.

All analyses use t_0 defined as the first stimulus onset of each trial, fixed by pipeline design before outcome inspection. This definition is not selected post hoc.

7 Proposed Experimental Validation: PTSP

7.1 Design Principle

Standardize and preregister the trigger; randomize or systematically manipulate the delay.

The target phase $\phi^* \in [0, 2\pi)$ is preregistered according to the target system. No specific value such as 180° is predicted here.

7.2 Target Event Preregistration

The definition of t_0 is a scientific commitment, not a free parameter. If researchers select the event definition producing the strongest observed relationship with learning, the hypothesis becomes unfalsifiable by construction. PTSP therefore requires that the operational definition of t_0 (signal, filter, detector, threshold, phase) be specified before outcome data are examined, justified by independent reasoning, and reported as an exploratory deviation if changed.

7.3 Latency Conditions

Three conceptual conditions:

- **Fast arm.** Low-latency pipeline, candidate range $L \in [1, 20]$ ms.
- **Slow arm.** Conventional pipeline, candidate range $L \in [50, 500]$ ms.
- **Programmable-delay arm.** A controlled delay stage samples intermediate gaps and permits direct manipulation of both latency and jitter.

These are proposed experimental regions, not biological thresholds.

7.4 Threats to Causal Identification

The central challenge is that fast and slow conditions may not deliver "the same feedback at different times." Participants may experience them as qualitatively different events. PTSP specifies the following controls:

1. randomized trial-level delay assignment;
2. identical feedback content across conditions;
3. identical feedback duration and intensity;
4. blinding where possible;

5. counterbalancing of order, fatigue, and practice;
6. analysis on achieved latency L_i, not nominal condition labels;
7. independent manipulation checks for perceived contingency and temporal predictability;
8. baseline block before the manipulation;
9. primary outcome measured after the intervention, not during it;
10. where feasible, temporal predictability manipulated as a separate factor with matched mean.

7.5 Power and Trial Allocation

The delay range should be sampled approximately uniformly on a log scale, with sufficient trials in the tail to distinguish competing distributions. A design concentrated near zero delay has almost no power to discriminate heavy-tailed from light-tailed alternatives. The primary analysis should specify a minimum effect size of theoretical interest in advance, so that a null result is interpretable as evidence rather than inconclusive.

7.6 Primary Outcome and Candidate Models

The primary dependent variable should be defined independently of the latency manipulation. A useful formulation is

$$Y_i = f(L_i, \mathbf{z}_i; \theta) + \epsilon_i, \tag{4}$$

where Y_i is the learning-related outcome, L_i the achieved gap, $\mathbf{z}_i$ prespecified covariates, and ϵ_i unexplained variation. The critical test is whether L_i explains out-of-sample variation in Y_i after controls.

If a functional form is fit, it must compete against alternatives. Candidate forms include

$$E_{\text{exp}}(L) = e^{-L/\tau}, \tag{5}$$

$$E_{\text{Gauss}}(L) = e^{-(L/\sigma)^2}, \tag{6}$$

$$E_{\text{pow}}(L) = \left(1 + \frac{L}{\tau}\right)^{-\alpha}, \tag{7}$$

$$E_{\text{L}}(L) = \frac{c}{L^2 + c}, \qquad c > 0. \tag{8}$$

No candidate has theoretical priority. The Lorentzian form is included because its heavy tail is one plausible shape, not because it is derived from biology. Model comparison should use held-out or cross-validated predictive criteria, and all fitted parameters should be reported with uncertainty.

8 Simulations and Their Proper Interpretation

Simulations of the proposed closed-loop structure can illustrate what the framework implies under specified assumptions. They cannot establish biological constants.

Any simulation reported in this work should state:

1. every parameter value and its source (specified, estimated, or arbitrary);
2. the sampling rate, buffer size, decay function, and random seed;

3. what happens to the output when each parameter is varied;

4. and, explicitly, that the output is a property of the simulation, not a measurement of the brain.

If a simulation generates synthetic data and then fits a candidate model to that data, the resulting fit reflects the assumptions used to generate the data, not the empirical adequacy of the model. Such exercises can be useful for checking that an inference procedure recovers known ground truth, but they are not evidence for a biological claim.

9 Scope and Limitations

This paper proposes a measurement and a testable prediction. It does not propose a biological mechanism, does not derive a functional form, and does not present simulation output as evidence.

- The predictive hypothesis and the distributional prediction are the only scientific claims. If they fail, the contribution reduces to a measurement convention.
- The causal hypothesis requires controlled manipulation and has not been tested here.
- The preliminary analysis uses a single dataset. Independent replication has not been performed.
- No kernel is derived from biology. If a kernel is fit empirically, it must compete against alternatives and be validated out of sample.
- No numerical time constant is claimed. Any characteristic scale that appears in a candidate model is a fitted parameter, not a biological constant.
- Biological eligibility traces may be multi-timescale, non-exponential, and state-dependent. The simple exponential form used for illustration is not a biological model.
- If potentiation- and depression-associated eligibility traces differ in their temporal persistence, this would predict an asymmetry in the effective Write-Back Gap. Testing this asymmetry is future work; no specific time constants are claimed here.
- Software implementations of these ideas are engineering artifacts, not evidence for the framework.
- Physiological state is multidimensional; no single measure should be assumed to summarize it.
- Feedback modality may interact with the timing effect in ways not captured by a scalar L.
- Causal identification is difficult because latency manipulation may inadvertently alter feedback quality, predictability, or workload.

10 Research Roadmap

The framework can be evaluated as a sequence of increasingly demanding tests.

1. **Measurement validation.** Establish test–retest reliability of L at trial resolution.
2. **Distributional characterization.** Quantify mean, variance, jitter, and tail behavior. (Addressed in Section 6.)

3. **Synthetic validation.** Stress-test the identification procedure using synthetic datasets with known ground-truth kernels, to confirm that the analysis can recover a known answer before it is applied to real data.

4. **Controlled manipulation.** Vary L while holding feedback content constant, following PTSP.

5. **Model competition.** Compare candidate functions on held-out data with adequate tail sampling.

6. **Distributional test.** Test whether variance or tail statistics predict outcome after controlling for the mean. (Supported in Section 6.)

7. **Independent replication.** Estimate parameters in independent samples and contexts.

8. **Engineering translation.** Only after biological timing effects are supported, evaluate whether architectures improving the relevant distribution produce measurable benefits.

11 Conclusion

The Write-Back Gap is proposed as a common temporal variable for studying closed-loop adaptive systems. The central idea is that feedback does not occur in a temporal vacuum: between a target biological event and the delivery of an adaptive intervention, the biological state relevant to learning may change.

The strongest claim in this paper is not that a particular latency is optimal, nor that any specific functional form is correct. It is the more general and testable proposition that the temporal distribution of feedback may contain information about learning outcomes that is lost when systems are described only by task-level performance or average latency.

A survey of public datasets found that trial-level feedback timing is not routinely recorded across the field, but is preserved in bigP3BCI Study M. Preliminary analysis of 444 trials from 15 subjects supports the predictive hypothesis and supports the distributional prediction: the variance of L carries information about accuracy beyond its mean. The causal hypothesis remains untested, and independent replication has not been performed. The next step is for other laboratories to record the events.

Whether the proposition survives is an empirical question. The contribution of this paper is to state it precisely enough to be tested, to document that the data is not routinely published, and to specify the protocol by which the causal claim could be falsified.

> *A closed-loop learning system should be evaluated not merely by what feedback it provides, but by when, how consistently, and under what biological conditions that feedback is received.*

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